INCH-POUND

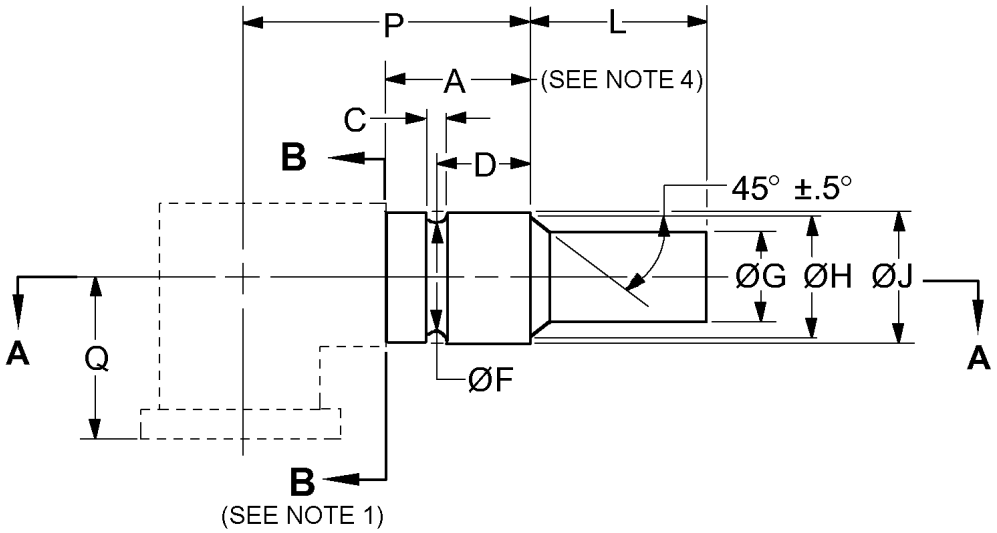
MS27082E
w/AMENDMENT 1
31 May 2017
SUPERSEDING
MS27082E
14 January 2014

DETAIL SPECIFICATION SHEET

ELBOW, NIPPLE END, SWIVEL FLANGE TO HOSE - 90°

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

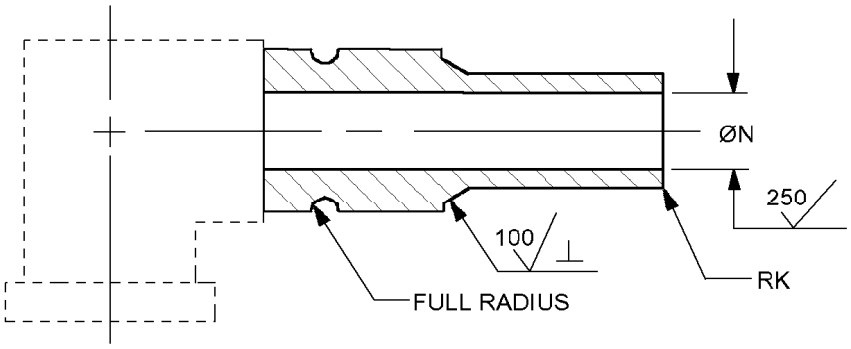
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27272.



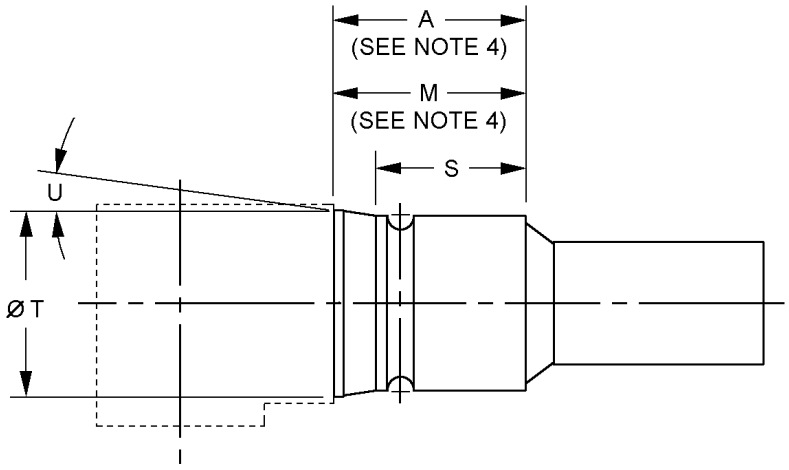
See notes at end of figure.

FIGURE 1. Elbow illustration.

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SECTION A-A



-12 THRU -24 SIZE ONLY

Size and material code		A min (see note 4)	C		D + .005 -.000	
Corrosion resistant steel (CRES)	Aluminum (Al)				CRES	Al
-8C	-8D	.617	.098	+.004 -.000	.385	.385
-10C	-10D	.654			.420	.427
-12C	-12D	.755	.128	+.005 -.000	.500	.500
-16C	-16D	.831			.545	.545
-20C	-20D	.881			.565	.571
-24C	-24D	1.035			.665	.665

See notes at end of figure.

FIGURE 1. Elbow illustration - Continued.

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Size and material code		F dia.		G dia. +.005 -.000	H dia. ±.005	J dia. +.005 -.000	K	
CRES	Al							
-8C	-8D	.497	+.005 -.000	.426	.530	.616	.020	+.005 -.000
-10C	-10D	.586		.526	.625	.706		
-12C	-12D	.674		.650	.760	.826	.030	±.005
-16C	-16D	1.001	+.008 -.000	.900	1.040	1.150		
-20C	-20D	1.255	+.005	1.151	1.275	1.405		
-24C	-24D	1.490	-.000	1.401	1.550	1.635	.035	

Size and material code		L		M (see note 4)	N dia.		
CRES	Al				CRES	Al	
-8C	-8D	.600	±.015	.583	.345	.345	+.006 -.000
-10C	-10D	.650		.620	.440	.440	
-12C	-12D	.675	±.025	.720	.560	.560	
-16C	-16D	.730		.796	.828	.828	
-20C	-20D	.935		.846	1.058	1.058	
-24C	-24D	.980		1.000	1.253	1.282	+.005 -.000

Size and material code		P ±.035	Q ±.035	S ±.010	T dia. max	U max
CRES	Al					
-8C	-8D	1.135	.892	---	---	---
-10C	-10D	1.315	.896	---	---	---
-12C	-12D	1.808	1.156	.625	.900	15°30´
-16C	-16D	1.901	1.282	.670	1.190	10°30´
-20C	-20D	2.180	1.500	.695	1.485	15°30´
-24C	-24D	2.500	1.688	.795	1.750	

NOTES:

1. Any bent tube design of the elbow to the left of plane B-B is acceptable provided the dimensions of P and Q and the requirements of this specification sheet and the procurement specification are met.
2. The inside diameter (ID) of the elbow shall not be less than the ID of the nipple end of the elbow.
3. Ovality shall not exceed 7.5 percent of the nominal tubing outside diameter (OD).
4. Use A dimension when the adjacent diameter to the left of plane B-B is greater than J dimension. When the adjacent diameter is equal to or less than J dimension, M dimension may be used in place of A dimension.
5. Surface roughness. Unless otherwise specified, maximum surface roughness shall not exceed 125 μin. R_a in accordance with ASME B46.1.
6. Unless otherwise specified, break or radius all corners .005, +.005, -.000.
7. All diameters within length A plus L must be concentric within .005 full indicator movement.
8. Remove all burrs and slivers.

FIGURE 1. Elbow illustration - Continued.

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REQUIREMENTS

Elbow illustration. See figure 1.

Intended use. This part is a component of MS27066. This is a design standard for manufacturing purposes. The item is only procured as an integral part of adapter assemblies.

In house manufacturing qualification: Qualified sources for MS27066 that meet the qualification testing requirements in MIL-DTL-27272 require no further testing for the 90° elbow.

Subcontracted manufacturers qualification of 90° elbow only: The 90° elbow shall be installed on a qualified connector and shall be subjected to the tests below in accordance with MIL-DTL-27272, reference periodic P1 and P2 testing in MIL-DTL-27272. Sample size, if qualifying all connector configurations, shall be performed on six fittings (three assemblies) for each size.

Examination of product
Proof pressure
Leakage pressure
Room-temperature burst pressure
Over-tightening torque

Materials. Material and material codes see table I.

TABLE I. Materials and material codes.

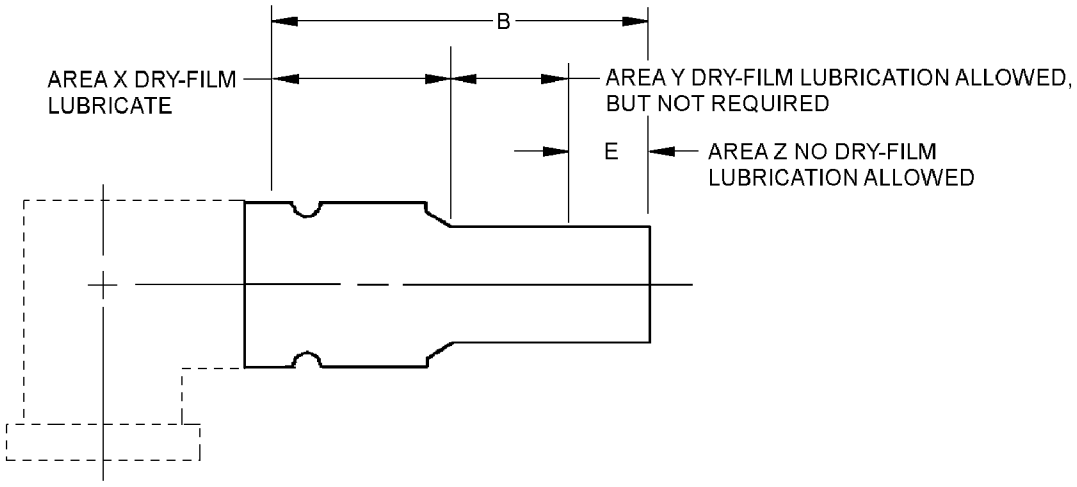
Material code	Material
C	Corrosion-resistant steel (CRES), class 304 or 321 cold drawn or cold rolled in accordance with SAE-AMS-QQ-S-763, SAE-AMS5639, or SAE-AMS5645.
D	Aluminum alloy, 6061-T6 or T651 in accordance with SAE-AMS-QQ-A-367, SAE-AMS4117 or alloy 7075-T73, 7075-T7351 in accordance with SAE-AMS-QQ-A-225/9 or alloy 7055-T7351 in accordance with SAE-AMS4124.

Finish. Corrosion-resistant steel, passivate in accordance with SAE-AMS-2700, method 1, type 6 or 7. Dry-film lubricate area X, see figure 2 and table II.

Aluminum alloy. Anodize in accordance with MIL-A-8625, type II, dye blue.

NOTE: Avoid using graphite dry film lubes with aluminum nipples because in a wet environment, graphite becomes corrosive to the aluminum.

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Size and material code		B min	E	
CRES	Al			
-8C	- 8D	1.030	.32	±.12
-10C	-10D	1.130	.35	±.15
-12C	-12D	1.240		
-16C	-16D	1.340	.39	±.19
-20C	-20D	1.570	.48	±.28
-24C	-24D	1.720	.50	±.30

FIGURE 2. Dry-film lubrication area.

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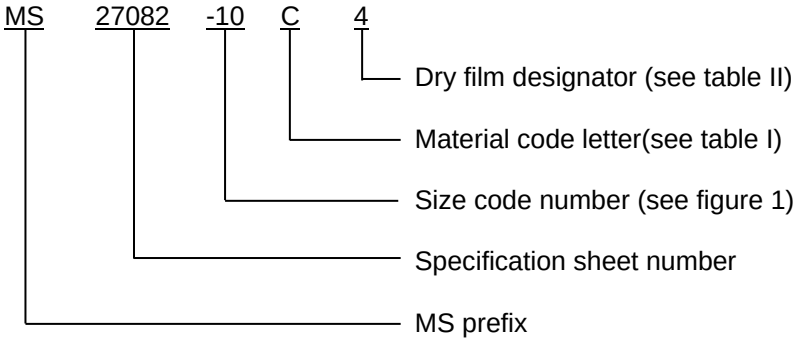
TABLE II. Dry film designator.

Dry film designator	SAE class or type designator	Dry film characteristics
Blank	Any SAE class or type below	N/A
SAE-AS1701	SAE-AS1701 class	SAE-AS1701 temperature ranges °F (°C)
4	4	-65° to +1400°F (-54° to 760°C)
5	5	-65° to +850°F (-54° to 454°C)
6	6	-375° to +850°F (-226° to 454°C)
SAE-AS5272	SAE-AS5272 type	SAE-AS5272 temperature ranges. °F (°C)
7	Type I	-90° to 400°F (-68 to 204°C) endurance life of 250 min minimum
8	Type II	-90° to 400°F (-68° to 204°C) endurance life of 450 min minimum
9	Type III	Color 1 - Natural product color -90° to 400°F (-68 to 204°C) low Volatile organic compound with an endurance life of 450 min minimum
10	Type III	Color 2 - Black color -90° to 400°F (-68 to 204°C) low Volatile organic compound with an endurance life of 450 min minimum
Dry film designator	MIL classification	Dry film characteristics
MIL-PRF-46010 1/	---	MIL-PRF-46010 temperature ranges. °F (°C)
11	1	Color 1 natural product color, -90° to 400°F (-68 to 204°C) solvent resisting
12	2	Color 2 - Black color -90° to 400°F (-68 to 204°C) solvent resisting

1/ Not for aerospace usage.

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PIN: The PIN consists of prefix “MS”, the specification sheet number, dash number for 90° elbow size, letter for material, and a blank or number for dry film lubricant. Unassigned PIN’s shall not be used.



PIN examples:

MS27082-10D	Indicates a 90° elbow size 10, aluminum and dry film class designator “blank”.
MS27082-10C	Indicates a 90° elbow size 10, CRES and dry film class designator “blank”.
MS27082-10C4	Indicates a 90° elbow size 10, CRES with dry film class designator 4.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-27272, this document references the following:

MIL-PRF-46010	SAE-AMS-QQ-A-367	SAE-AMS5639
MIL-A-8625	SAE-AMS-QQ-S-763	SAE-AMS5645
MS27066	SAE-AMS2700	SAE-AS1701
ASME B46.1	SAE-AMS4117	SAE-AS5272
SAE-AMS-QQ-A-225/9	SAE-AMS4124	

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

Project 4730-2017-069)

Review activities:
Army - AR, AT, MI
Navy - MC, SA, SH
Air Force - 71

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